RIYANSH VERMA

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SUMMARY

Software Engineer with full-stack development experience across React, TypeScript, Node.js, and FastAPI, backed by PostgreSQL and Supabase, gained through internships and independent projects. Designs and ships REST APIs, authentication systems, and relational database schemas end-to-end. Works as an AI engineering specialist as a secondary focus, building Retrieval-Augmented Generation pipelines and multi-agent LLM systems with LangChain and LangGraph for production chatbot and assessment platforms.

TECHNICAL SKILLS

Languages: TypeScript, JavaScript, Python, SQL, C++

Frontend: React, Next.js, Redux Toolkit, Zustand, Tailwind CSS, shadcn/ui, Radix UI, Chart.js

Backend: Node.js, Bun, Express, Elysia, Hono, FastAPI, REST APIs, WebSockets

Databases: PostgreSQL, MySQL, Sqlite, MongoDB, Redis, ChromaDB, CockroachDB

AI Engineering: LangChain, LangGraph, RAG, LLM Applications, AI Agents, Embeddings, Vector Search, Prompt Engineering

Developer Tools: Git, GitHub, Docker, Postman, Claude, ChatGPT

Engineering Concepts: Data Structures & Algorithms, OOPs, Database Design, API Design, System Design, Operating System, Computer Networks, Microservices, Message Queues, Event-Driven Architecture , WebHooks

Testing & Validation: PyTest, Playwright, Zod, Pydantic, Vitest, Unit Testing, Integrated Testing, E2E Testing

EXPERIENCE

Full Stack Developer Intern | Dynamicore Strategies *Jaipur | May 2026 – Aug 2026*

- Engineered a multi-agent financial assistant using Python, LangGraph, and LangChain, orchestrating specialist agents and 15+ tools for portfolio management, investment guidance, and financial education.
- Built a RAG pipeline over 60,000+ finance and education PDFs using hybrid search, CockroachDB, PostgreSQL, Redis caching, rate limiting, and multi-layer memory; developed LoRA fine-tuned guardrails on 1M+ prompts with LangSmith, LangServe, Pydantic, and PyTest, achieving ∼200ms classification and 4–5s TTFT.

Full Stack Engineer Intern | The Learner's Academy *Jaipur | Jan 2026 – May 2026*

- Engineered an end-to-end school ERP for ∼500 users with RBAC across admin, teacher, student, parent, and employee workflows using React, Bun, Elysia, and PostgreSQL.
- Built RESTful APIs with caching and rate limiting; integrated Razorpay payments, webhooks, and WebRTC for real-time parent–teacher meetings, with type-safe forms via React Hook Form/Zod, Vitest testing, Turborepo, and Vercel, Render, and Supabase deployment.

Front-End Developer Intern | Dynamicore Strategies *Jaipur | May 2025 – June 2025*

- Built a fintech platform frontend across 8 technologies, covering real-time portfolio tracking, an investment explorer, live market data, loan comparison tools, KYC integration flows, and gamified financial-literacy learning modules using React, TypeScript, and Chart.js.
- Implemented a responsive, mobile-first UI using shadcn/ui, Radix UI, and Tailwind CSS.

Data Visualization Intern | Aunwesha Knowledge Tech *Kolkata | May 2024 – June 2024*

- Analyzed Kaggle datasets and built interactive dashboards using Tableau, alongside Java-based CRUD operations against MySQL via JDBC for dataset management.
- Built data-visualization web tools using Vega-Lite and jQuery/AJAX across 4 chart types (bar, line, pie, time-series).

PROJECTS

Trajectory | AI-Powered Career Guidance System *Live | Code*

- Trained an MLP psychometric model on 1M+ Big Five and 750K+ RIASEC response sets, achieving 99.98% accuracy; scored 91-question assessments to derive personality traits and map users to suitable career roadmaps.

- Engineered a microservices platform with React, Bun/Express, PostgreSQL, Redux, and REST APIs, integrating a FastAPI RAG service with ChromaDB for personalized roadmaps, with Zod/Pydantic validation and rate limiting.

Suvidha | AI-Powered Healthcare Ecosystem *Code*

- Engineered an AI healthcare platform with React, Bun/Hono, PostgreSQL, Redis, and BullMQ, implementing priority queues, 1-min waitlist allocation, real-time notifications, access controls, webhooks, and event-driven scheduling.
- Built multimodal AI for medical image analysis, TTS/STT, health insights, and WhatsApp assistance using 250K+ medicine records; implemented caching, rate limiting, streaming, and 10 compliance controls for observability.

4-in-a-Row | Real-Time Multiplayer Game *Code*

- Engineered a real-time multiplayer game using React, Bun/Fastify, Socket.io, Redis, and PostgreSQL, implementing matchmaking, reconnection, move timers, persistent history, leaderboards, and JWT authentication.
- Built a TypeScript game engine with Minimax, alpha-beta pruning, and bitboards; integrated a Python/FastAPI AI assistant with Pydantic, Ollama, REST/SSE, and Redis for grounded, streaming hints.

Ryder | Car Rental Platform *Live* | *Code*

- Engineered a location-based car rental platform using React, Node.js, Express, PostgreSQL, Redux, for nearby car discovery, price comparison, owner fleet browsing, trip-fare monitoring, and booking with OTP-verification.
- Built workflows with Razorpay payments, and fleet management, enabling vehicle listings, availability and maintenance tracking, revenue and popular-car analytics, and automatic vehicle-model generation from specifications.

TurfOnTop | Turf Booking Platform *Live* | *Code*

- Engineered a location-aware turf booking platform using React, Node.js, Express, PostgreSQL, Redux, and Google Maps for nearby turf discovery by distance, travel time, traffic, and weather, with slot booking, cancellation, and online payments.
- Built owner workflows and analytics dashboards for turf listings, operating hours, capacity, booking management, customer follow-ups, revenue, and peak-hour analysis, integrating Razorpay and Cloudinary.

EDUCATION

B.Tech in Computer Science | JK Lakshmipat University, Rajasthan 2022–2026 | CGPA: 7.2/10

High School | Delhi Public School, Rajasthan 2021–2022 | 88%

RESEARCH & PUBLICATIONS

Decentralized Carbon Markets: Blockchain-Enabled Carbon Credit Exchange | IEEE | *Published April 2026*

- Co-authored research addressing inefficiency, lack of transparency, and limited accessibility in offline carbon credit trading through a blockchain-based decentralized exchange design; available here.